

```
In [7]: import sys
        !{sys.executable} -m pip install -r cb_requirements.txt
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy>=1.24 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 1)) (2.1.3)
Requirement already satisfied: pandas>=2.0 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 2)) (2.2.3)
Requirement already satisfied: scikit-learn>=1.3 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 3)) (1.6.1)
Requirement already satisfied: matplotlib>=3.7 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 4)) (3.10.0)
Requirement already satisfied: seaborn>=0.12 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 5)) (0.13.2)
Requirement already satisfied: scipy>=1.11 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 6)) (1.15.3)
Requirement already satisfied: joblib>=1.3 in c:\programdata\anaconda3\lib\site-packages (from -r cb_requirements.txt (line 7)) (1.4.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\lib\site-packages (from pandas>=2.0->-r cb_requirements.txt (line 2)) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=2.0->-r cb_requirements.txt (line 2)) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas>=2.0->-r cb_requirements.txt (line 2)) (2025.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn>=1.3->-r cb_requirements.txt (line 3)) (3.5.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (1.3.1)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (4.55.3)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (1.4.8)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (24.2)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib>=3.7->-r cb_requirements.txt (line 4)) (3.2.0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas>=2.0->-r cb_requirements.txt (line 2)) (1.17.0)

```
In [8]: %matplotlib inline
```

```
In [9]: %run cb_main.py
```

```
=====
CYBERBULLYING DETECTION ON SOCIAL MEDIA – NLP PROJECT
=====
```

[STEP 1] Generating social media dataset...

```
Shape : (10000, 3)
Labels :
label
normal      4500
toxic       1800
obscene     1200
insult      1000
identity_hate 700
severe_toxic 500
threat      300
```

[STEP 2] Extracting handcrafted text features...

Features added: char_count, word_count, avg_word_len, exclamation_count, caps_ratio, unique_word_ratio

[STEP 3] Cleaning and preprocessing text (NLP pipeline)...

```
Done in 0.4s
Original : fantastic kind opinion in this comment excited support amazing good
point good point thanks
Cleaned  : fantastic kind opinion comment excited support amazing good point g
ood point thanks
```

[STEP 4] Encoding labels and splitting dataset (80/20)...

```
Classes: ['identity_hate', 'insult', 'normal', 'obscene', 'severe_toxic', 'thre
at', 'toxic']
Train: 8000 | Test: 2000
```

[STEP 5] TF-IDF Feature Extraction (unigrams + bigrams)...

```
Vocabulary size : 6,642
Train matrix    : (8000, 6642)
Test matrix     : (2000, 6642)
```

[STEP 6] Training all models...

```
[1/4] Naive Bayes (Complement NB)...
done 0.01s
[2/4] Logistic Regression...
done 15.61s
[3/4] Linear SVM...
done 0.49s
[4/4] Random Forest...
done 0.81s
```

[STEP 7] 5-Fold Stratified Cross Validation...

```
Naive Bayes      F1=0.8474 +- 0.0110
Logistic Regression F1=0.9079 +- 0.0109
Linear SVM       F1=0.9187 +- 0.0054
Random Forest    F1=0.9115 +- 0.0110
```

[EVALUATING ALL MODELS]

```
=====
CYBERBULLYING DETECTION – FINAL RESULTS
=====
Model          Accuracy F1-Macro Precision Recall ROC-AUC
-----
```

Naive Bayes	0.8945	0.8237	1.0000	1.0000	1.0000
Logistic Regression	0.9375	0.9017	1.0000	1.0000	1.0000
Linear SVM	0.9380	0.9145	0.9991	1.0000	0.9994
Random Forest	0.9385	0.9080	1.0000	1.0000	1.0000

=====

Best Model (F1-Macro): Linear SVM (0.9145)

=====

Detailed Classification Reports:

[Naive Bayes]

	precision	recall	f1-score	support
identity_hate	1.00	1.00	1.00	126
insult	0.92	1.00	0.96	188
normal	1.00	1.00	1.00	900
obscene	0.88	0.47	0.61	254
severe_toxic	0.98	0.43	0.59	108
threat	0.66	0.98	0.79	55
toxic	0.70	0.96	0.81	369
accuracy			0.89	2000
macro avg	0.88	0.83	0.82	2000
weighted avg	0.91	0.89	0.88	2000

[Logistic Regression]

	precision	recall	f1-score	support
identity_hate	1.00	1.00	1.00	126
insult	0.98	1.00	0.99	188
normal	1.00	1.00	1.00	900
obscene	0.80	0.82	0.81	254
severe_toxic	0.97	0.66	0.78	108
threat	0.73	1.00	0.85	55
toxic	0.87	0.88	0.88	369
accuracy			0.94	2000
macro avg	0.91	0.91	0.90	2000
weighted avg	0.94	0.94	0.94	2000

[Linear SVM]

	precision	recall	f1-score	support
identity_hate	1.00	1.00	1.00	126
insult	0.99	1.00	0.99	188
normal	1.00	1.00	1.00	900
obscene	0.82	0.77	0.79	254
severe_toxic	0.90	0.77	0.83	108
threat	0.86	0.98	0.92	55
toxic	0.85	0.90	0.87	369
accuracy			0.94	2000
macro avg	0.92	0.92	0.91	2000
weighted avg	0.94	0.94	0.94	2000

[Random Forest]

	precision	recall	f1-score	support
identity_hate	1.00	1.00	1.00	126
insult	0.98	0.99	0.99	188
normal	1.00	1.00	1.00	900
obscene	0.79	0.82	0.80	254
severe_toxic	1.00	0.69	0.82	108
threat	0.76	1.00	0.87	55
toxic	0.87	0.89	0.88	369
accuracy			0.94	2000
macro avg	0.92	0.91	0.91	2000
weighted avg	0.94	0.94	0.94	2000

[STEP 8] Saving visualisations...

9 plots saved to ./plots/

[STEP 9] Saving models...

Saved to ./models/

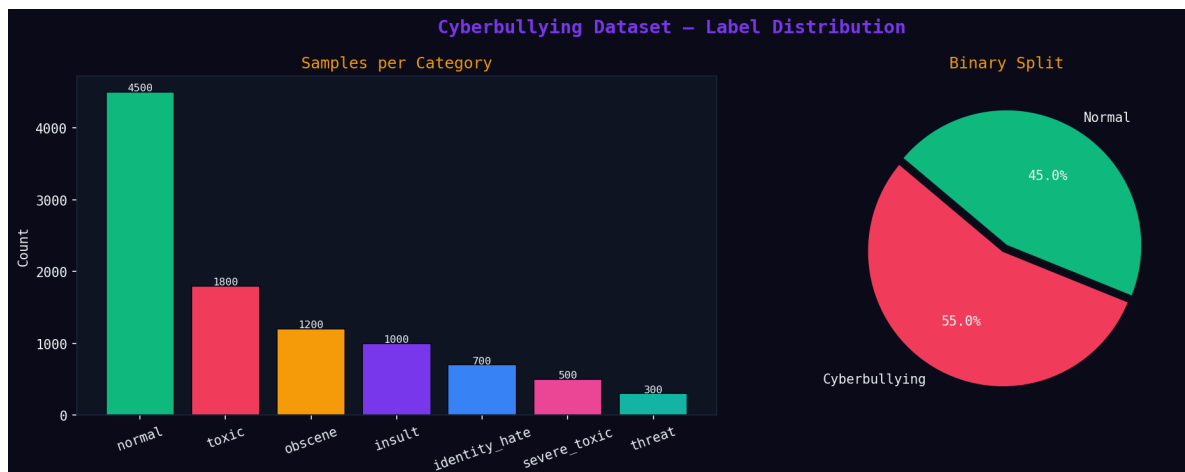
Done!

```
In [13]: from IPython.display import Image, display
```

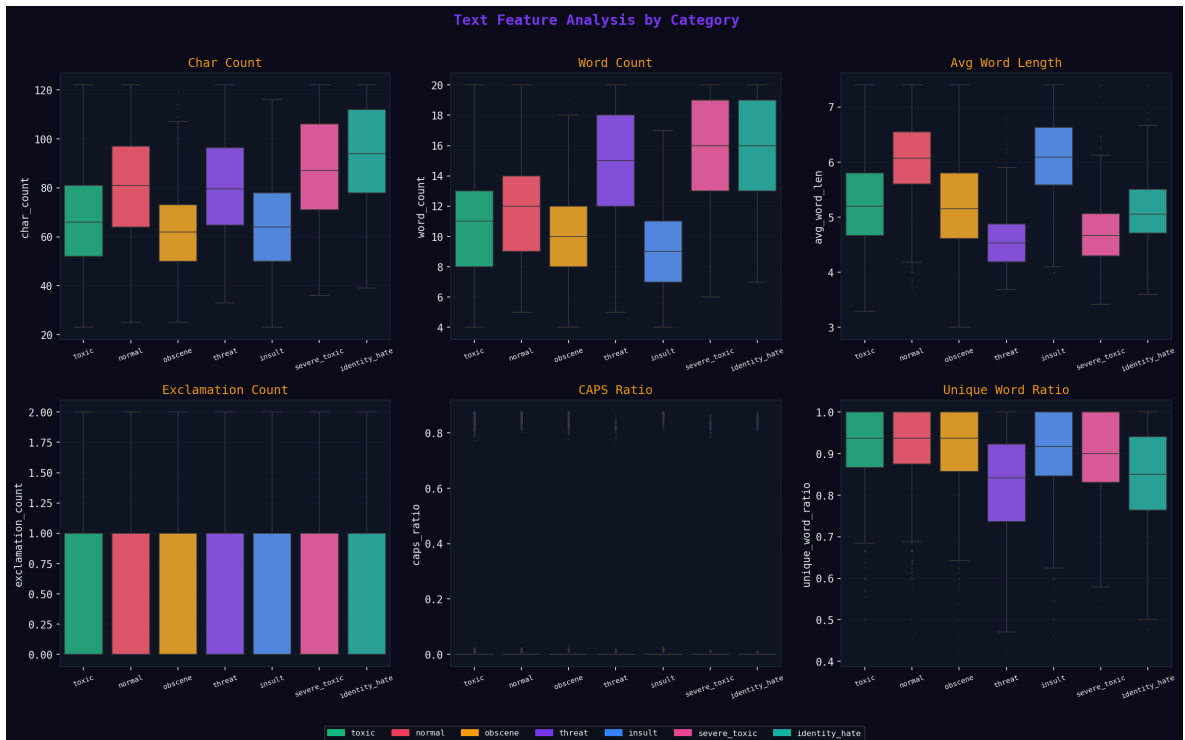
```
plots = [
    'plots/01_label_distribution.png',
    'plots/02_text_feature_analysis.png',
    'plots/03_model_comparison.png',
    'plots/04_confusion_matrices.png',
    'plots/05_top_tfidf_words.png',
    'plots/06_roc_curves.png',
    'plots/07_word_frequency.png',
    'plots/08_cross_validation.png',
    'plots/09_per_class_f1.png'
]

for p in plots:
    print(f"\n{' '*50}\n{p}\n{' '*50}")
    display(Image(p))
```

plots/01_label_distribution.png



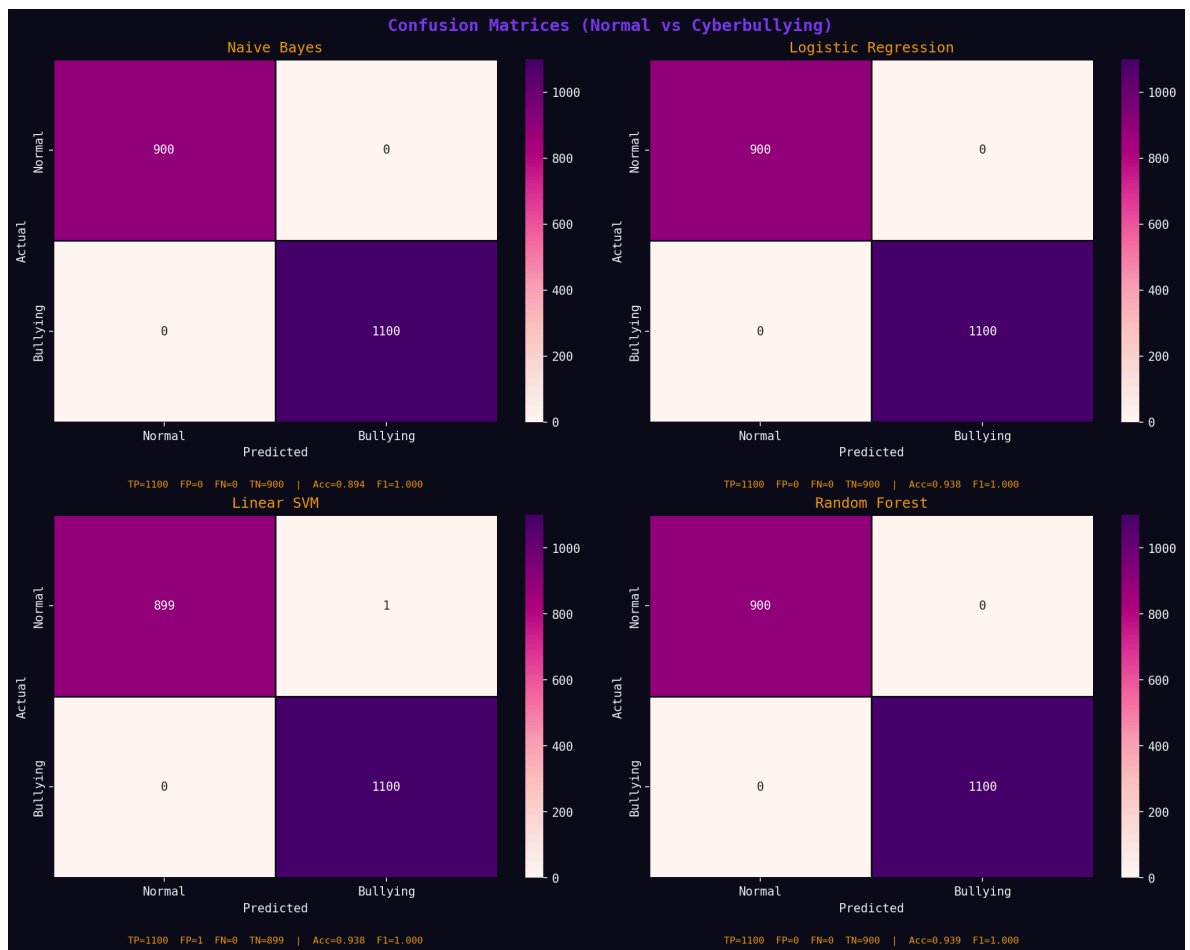
plots/02_text_feature_analysis.png



plots/03_model_comparison.png



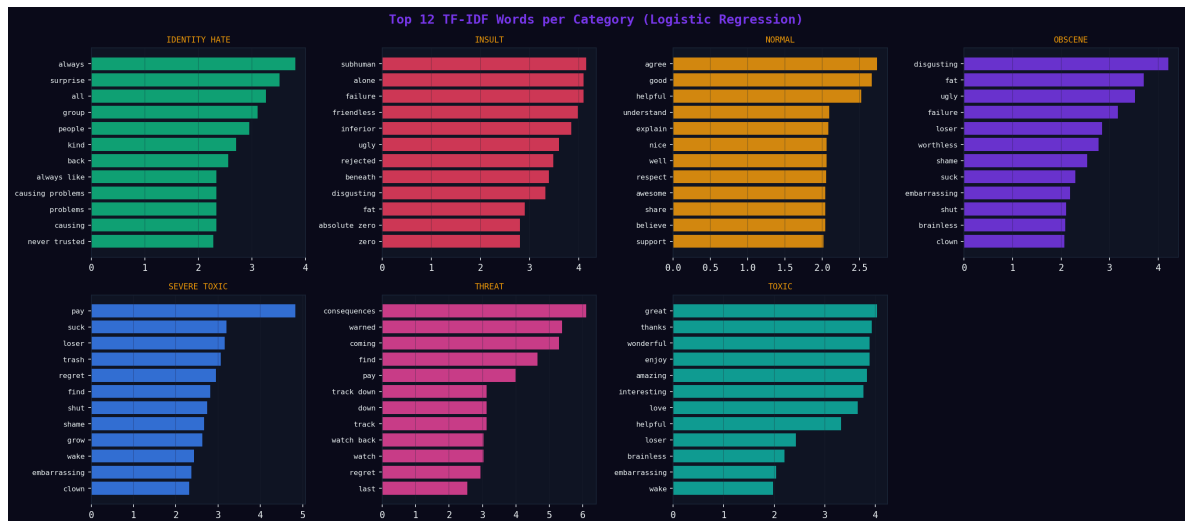
plots/04_confusion_matrices.png



=====

plots/05_top_tfidf_words.png

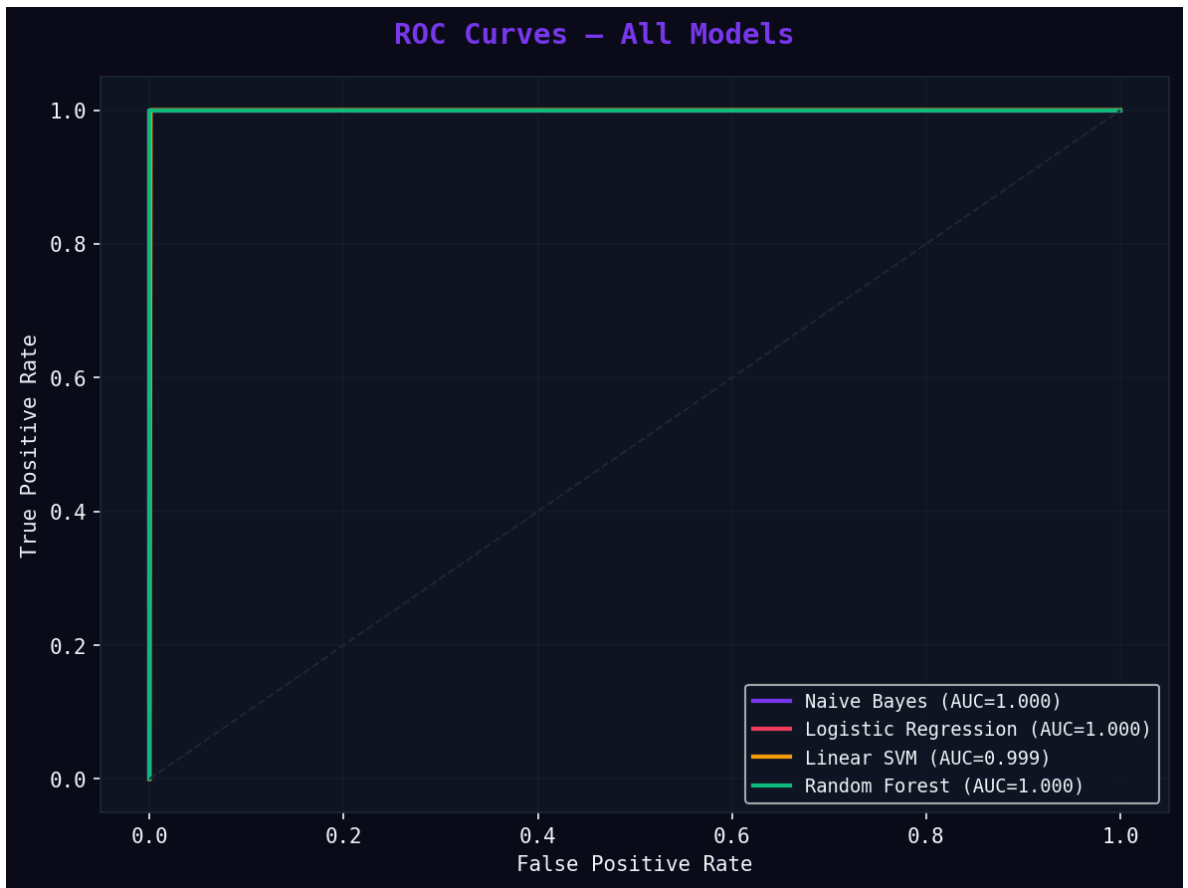
=====



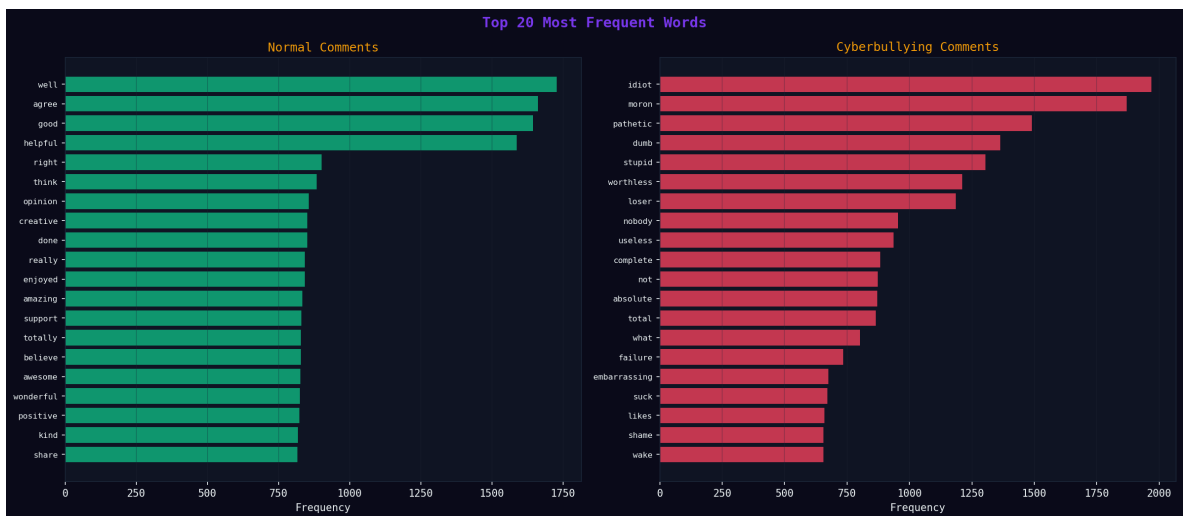
=====

plots/06_roc_curves.png

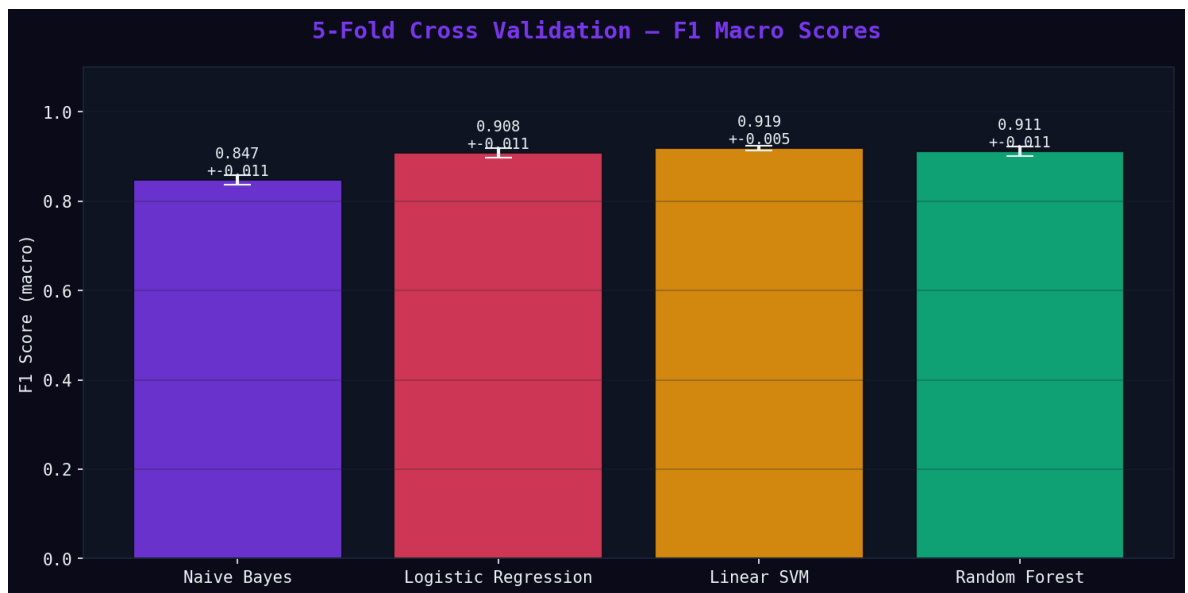
=====



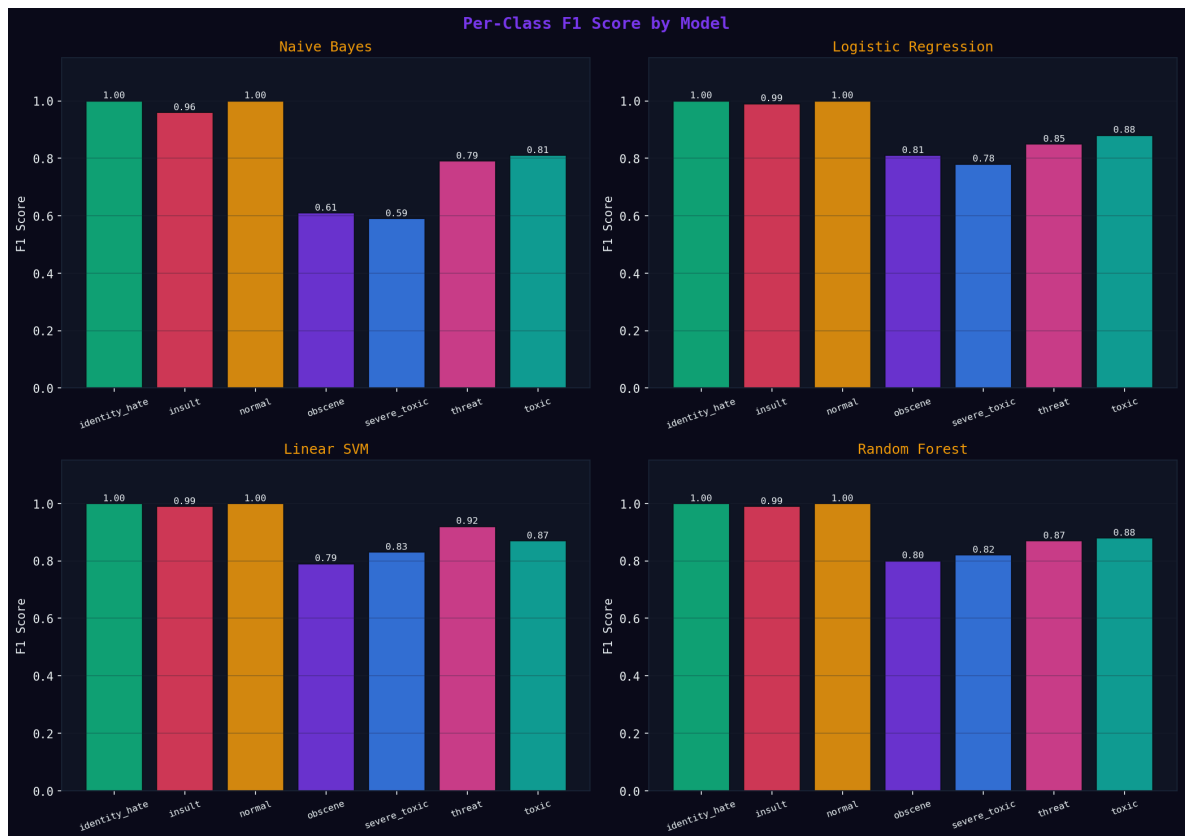
plots/07_word_frequency.png



plots/08_cross_validation.png



plots/09_per_class_f1.png



In [23]: `%run cb_eda.py`

Generating data for EDA...

– DATASET OVERVIEW –

```
Shape      : (8000, 13)
Columns    : ['text', 'label', 'is_cyberbullying', 'char_count', 'word_count',
'avg_word_len', 'exclamation_count', 'caps_ratio', 'punct_count', 'unique_word_ratio', 'avg_word_length', 'text_length', 'cleaned_text']
Memory usage : 0.79 MB
Missing vals : 0
```

– LABEL DISTRIBUTION –

```
label
normal      3600
toxic       1440
obscene      960
insult       800
identity_hate 560
severe_toxic 400
threat       240
Name: count, dtype: int64
```

– CYBERBULLYING SPLIT –

```
is_cyberbullying
1      4400
0      3600
Name: count, dtype: int64
```

– SAMPLE DATA –

```
                                text \
0  happy looking forward understand good point in...
1  tbh learn nice fantastic nice fantastic hope k...
2  excellent respect love fantastic thanks love f...
```

```
                                cleaned_text
0  happy looking forward understand good point in...
1  honestly learn nice fantastic nice fantastic h...
2  excellent respect love fantastic thanks love f...
```

[EDA] Generating plots...

```
Saved: plots/eda\eda_01_category_stats_heatmap.png
Saved: plots/eda\eda_02_text_length_violin.png
Saved: plots/eda\eda_03_bigrams.png
Saved: plots/eda\eda_04_feature_correlation.png
Saved: plots/eda\eda_05_aggression_signals.png
```

EDA complete! Plots saved to ./plots/eda/

In [25]: %run cb_inference.py

[Loading models...]
All models loaded!

```
=====
REAL-TIME CYBERBULLYING DETECTION DEMO
=====
# Prediction          Conf Text Preview
-----
1 [ NORMAL          ]      0% This is such a great video, I reall...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
2 [CYBERBULLYING]     50% You are so stupid, nobody wants to ...
  Votes: Naive:norma Logistic:insul Linear:insul Random:norma
3 [CYBERBULLYING]     75% Thanks for sharing this helpful inf...
  Votes: Naive:norma Logistic:toxic Linear:toxic Random:toxic
4 [ NORMAL          ]      0% I hope something terrible happens t...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
5 [ NORMAL          ]      0% Great explanation, very easy to und...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
6 [CYBERBULLYING]    100% You are the dumbest person I have e...
  Votes: Naive:toxic Logistic:toxic Linear:toxic Random:obsce
7 [ NORMAL          ]      0% This made my day, absolutely wonder...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
8 [CYBERBULLYING]    100% People like you should not be allow...
  Votes: Naive:ident Logistic:ident Linear:ident Random:ident
9 [ NORMAL          ]      0% I disagree with your point but I re...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
10 [CYBERBULLYING]   100% Say that again and see what happens...
  Votes: Naive:sever Logistic:threa Linear:sever Random:threa
11 [ NORMAL          ]      0% Really appreciate the effort you pu...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
12 [CYBERBULLYING]   100% What an idiot, completely brainless...
  Votes: Naive:toxic Logistic:toxic Linear:toxic Random:toxic
13 [ NORMAL          ]      0% Such a positive community, glad to ...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
14 [CYBERBULLYING]   50% You deserve every bad thing that co...
  Votes: Naive:ident Logistic:norma Linear:norma Random:obsce
15 [ NORMAL          ]      0% Looking forward to more content lik...
  Votes: Naive:norma Logistic:norma Linear:norma Random:norma
-----

Accuracy: 86.7% | Correct: 13/15
=====
```

[INTERACTIVE MODE] Type a comment to classify (or 'quit' to exit):

```
-----
Result      : CYBERBULLYING DETECTED
Confidence  : 50%
Naive Bayes      : identity_hate
Logistic Regression : normal
Linear SVM       : normal
Random Forest    : obscene
```

```
Result      : CYBERBULLYING DETECTED
Confidence  : 50%
Naive Bayes      : identity_hate
Logistic Regression : normal
Linear SVM       : normal
Random Forest    : obscene
```

```
Result      : Normal - No threat
Confidence  : 0%
Naive Bayes      : normal
Logistic Regression : normal
Linear SVM       : normal
Random Forest    : normal
```

```
Result      : CYBERBULLYING DETECTED
Confidence  : 50%
Naive Bayes      : identity_hate
Logistic Regression : normal
Linear SVM       : normal
Random Forest    : obscene
```

```
Result      : CYBERBULLYING DETECTED
Confidence  : 50%
Naive Bayes      : identity_hate
Logistic Regression : normal
Linear SVM       : normal
Random Forest    : obscene
```

```
Result      : CYBERBULLYING DETECTED
Confidence  : 75%
Naive Bayes      : normal
Logistic Regression : toxic
Linear SVM       : toxic
Random Forest    : toxic
```

Exiting demo.

In []: